Basic cyber security tricks

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Introduction – Cyber Security

- **Why we need cyber security?**

- **Cyber security** is the practice of defending computers, servers, mobile devices, electronic systems, networks, and data from malicious attacks. It's also known as information technology security or electronic information security.

- **CIA**

- **Vulnerability, Threat, Risk**: A vulnerability is a flaw or weakness in an asset's design, implementation, or operation and management that could be exploited by a threat.
- A threat is a potential for a threat agent to exploit a vulnerability.
- A risk is the potential for loss when the threat happens.

- Risk=Threat × Vulnerability

Cyber Crime

- Cyberspace is a domain characterized by the use of electronics and the electromagnetic spectrum to store, modify, and exchange data via networked systems and associated physical infrastructures.

- Advancement of knowledge of individuals in field of cyber space, the frequency of cyber crimes has increased over the last decade.

- The term cyber crime can be defined as an act committed or omitted in violation of a law forbidding or commanding it and for which punishment is imposed upon conviction.

- There are many bodies who are working all over the world. In India, CERT-IN (Indian Computer Emergency Response Team), Ministry of Electronics & Information Technology issues advisories for preventing the cyber crimes on regular basis.

Cyber crimes

- **Cyber-Stalking :** is a crime in which the attacker without being detected harasses a victim using electronic communication, such as e-mail or instant messaging(IM), or messages posted to Web site or a discussion group.
- **Hacking:** is the gaining of access (wanted or unwanted) to a computer and viewing, copying, or creating data (leaving a trace) without the intention of destroying data or maliciously harming the computer.
- **E-Mail Spoofing:** is the creation of email messages with a forged sender address.
- **Bot Networks**: is a collection of compromised computers often referred to as “zombies” infected with malware that allows an attacker to control them.
- **Phishing**
- **Malware**
- **Credit/Debit Card Fraud**
- **Spam**
- **Defamation**
- **Cyber Terrorism**

Phishing

Common phishing messages include:

- Pretending to be a bank or government institution asking for personal details to be urgently verified
- Pretending to be a manager and asking employees to make payments into a fraudulent bank account
- Intercepting a business' email conversations with a supplier and asking for payments to be made into a different bank account
- Messages saying you've won a competition and need to provide bank details to access the prize
- Cyber criminals often add urgency to phishing messages to encourage you to act.
- If you click on a link in a phishing message, it will usually take you to a fake website. Viruses can then be downloaded onto your computer or mobile phone. A virus is malicious computer code that can corrupt systems, destroy data, or steal passwords.

Source: TSB Bank

How to prevent Phishing

- Check the sender details in the message as it could be a spoof email pretending to be from a genuine person. Look for spelling alterations in the name, email address or website domain name
- Check for grammar and spelling errors, which can indicate it’s not from a legitimate organisation
- If the message includes a link, hover over it, but don't click, so that you can see the preview of the site it’s sending you to
- A legitimate company will never ask for your National Insurance number or PIN by email or text message
- If it looks too good to be true, it probably is. Hackers will try to hook you in with irresistible offers
- Check your business bank accounts often so that you spot any unusual activity. If you've lost money or details have changed, report to bank authority.

Malware

- **Malware** is malicious software that can destroy, damage, or exploit computers or computer systems.
- There are various types of malware including:
- **Viruses**
- A virus is malicious software that attaches itself to a computer attachment or file. It stays dormant until the attachment or file is opened or used. It can then spread throughout your computer system causing damage and data loss.
- Viruses can be spread by scam email and text message attachments and internet and app downloads.
- **Signs that your computer has been infected by a virus include:**
- It runs slow or programs take a long time to open
- It regularly freezes or programs crash
- You get unexpected pop-up windows that encourage you to visit unusual websites
- Unknown programs start when you switch on your computer
- Emails that you haven't created are sent from your account
- Businesses should warm employees not to open suspicious attachments or install new software on company devices. They should be encouraged to check with team members if there’s any suspicion.

Source: TSB Bank

Malware

Worms

•A worm is a malicious software that replicates itself and spreads from computer to computer. Unlike viruses, worms do not need to be attached to a computer program to do damage. They work silently and infect the device without the user's knowledge.

•Worms can delete or change files, take up hard drive space and allow hackers to seize control of a computer and steal data.

•Worms can spread through vulnerable software as well as via links in scam emails and text messages.

•**To detect if a worm has infected your computer, these steps are recommended:**

•Check your hard drive space. As worms replicate, they take up free space on your computer

•If your computer is running slow or programs keep crashing, it could be due to a worm

•Check if any files are missing which a worm may have removed

•It’s important that businesses have up to date virus protection software. Don’t put off making updates to software such as Windows. These updates often stop newly discovered viruses or vulnerabilities.

Malware

Ransomware

•Cyber criminals use ransomware to lock a device or steal information. They then demand a ransom to restore access or return the information. Payment is usually demanded as a crypto currency such as Bitcoin.

•Ransomware can spread in various ways. Methods include clicking on malicious links in an email or using an infected USB flash drive.

There are various types of ransomware. They include:

•Cryptoware: A user is blocked from accessing their data and files. A ransom is then demanded to restore access

•Scareware: Fake software appears to be genuine anti-virus software. The hackers claim they have found problems and ask for a payment to fix them

•Doxware: Criminals steal sensitive personal data, such as photographs, and threaten to publish it online if a ransom is not paid

Signs of ransomware include:

•New file extensions are added to the end of your file names

•Your computer is locked with a message demanding a ransom

Advice: Keep up to date virus software and staff awareness are the key options to reduce the chance your business is impacted by ransomware.

Malware

- **Spyware**
- Spyware is malicious software that infiltrates your device and monitors your activity. Criminals can then steal your login, password, and credit card information.
- Spyware can be spread by clicking on links in scam emails, downloading fake software and clicking on pop-up windows. Spyware can also be packaged with real programs.

- **Signs that your computer has been infected by spyware include:**
- It runs slow or programs keep crashing
- It keeps running out of hard drive space
- You get pop-up windows encouraging you to visit unusual websites
- Anti-virus software is not working

Malware

Trojan horse viruses

- A trojan horse is code or software that looks legitimate but can take control of your computer.
- A “backdoor trojan” allows fraudsters to access your device. They can then steal, change, or delete data as well as upload more malware.
- A trojan horse can also contain other types of malware including ransomware and spyware.
- Trojans are often sent in emails promoting software that businesses might be interested in such as training courses. If you click on a link, it can download harmful software.

Signs that your computer has been infected by a trojan include:

- It runs unusually slow, or programs keep crashing
- It keeps running out of hard drive space
- You get pop-up windows encouraging you to visit spam websites

Intellectual Property Crimes

- Intellectual property (IP) theft is defined as theft of material that is copyrighted, the theft of trade secrets, and trademark violations etc. A copyright is the legal right of an author, publisher, composer, or other person who creates a work.

Cross-site Scripting

- Cross-site scripting (XSS) is an attack in which an attacker injects malicious executable scripts into the code of a trusted application or website. Attackers often initiate an XSS attack by sending a malicious link to a user and enticing the user to click it. The malicious link executes the attacker's chosen code on the user's system. As a result, the attacker can-

- Capturing the keystrokes of a user
- Redirecting a user to a malicious website
- Obtaining the cookie information of a user who is logged into a website.

- **Child Pornography & Adult Pornography**

- Child Pornography is a criminal offence and is defined as any visual depiction involving the use of a minor, or one appearing to be a minor, engaging in sexually explicit conduct.

Steps for Hacking



Platform for Ethical Hacking

- Kali Linux
- https://www.virtualbox.org/ (install virtual box)
- Osboxes.org (virtualbox vdi 64 bit etc) install Kali
- Default user id- osboxes
- Pwd- osboxes.org

Google Dork

Extra google database search

•MongoDB is **an open-source document-oriented database** that is designed to store a large scale of data and also allows you to work with that

•Ext:pdf mongodb

•Exploit-db.com

Spe**cific search**:

•Inurl:login

•Intitle:press

Other search engines:

•Duckduckgo

•Bin

Information Gathering

archive.org (Wayback machine)

- https://www.nplindia.in/

Phishtank.org (check fake website)

Whois.sc for Reverse ip lookup and other detail

HTTRACK: https://www.httrack.com/ (website copier)

NMAP

Wireshark (Protocol analyser)

shodan.io

Website watcher: Trackly tool is used to monitor the site for any change

Few Tools/Tricks for Hacking

Password Database: root@kali# Crunch 4 6 abcd12 –o/root/password.lst

Rubber Ducky: Rubber Ducky, digispark

Tools to create Rainbow table: rtgen, winrtgen

SAM file & Mimikatatz:

Hashkiller: decrypt the hash to get the password

Information Technology Act 2000

- IT Act 2000 is an Act of the Indian Parliament (No 21 of 2000) notified on October 17, 2000 to have its exhaustive law to deal with the technology in the field of information technology including e-commerce, e-governance, e-banking etc as well as penalties and punishments in the field of cyber crimes.

  - **Section 69A Power to issue directions for blocking public access of any information through any computer resource**.—(1) In the interest of sovereignty and integrity of India (ii) defence of India (iii)security of the State (iv) friendly relations with foreign States (v) public order or (vi)for preventing incitement to the commission of any cognizable offence relating to above.

- The intermediary who fails to comply with the direction issued under sub-section (1) shall be punished with an imprisonment for a term which may extend to seven years and shall also be liable to fine.

- **Sec.66B:** Punishment for dishonestly receiving stolen computer resource or communication device.
- **Sec.66C:** Punishment for identity theft.
- **Sec.66E:** Punishment for violation of privacy.
- **Sec.66F** Cyber terrorism Imprisonment
- **Sec.67** Punishment for publishing or transmitting obscene material in electronic form.
- **Sec.67A** Punishment for publishing or transmitting of material containing sexually explicit act, etc., in electronic form
- **Sec.67B** Punishment for publishing or transmitting or storing of material depicting children in sexually explicit act, etc., in electronic form.
- **Sec.67C** Preservation and retention of information by intermediaries.

- CERT-IN (Indian Computer Emergency Response Team), Ministry of Electronics & Information Technology issues advisories on regular basis for common users on Desktop Security, Mobile Phone Security, Broadband Internet Security, USB Devices Security, Secure uses of Debit/Credit Card and preventing phishing attacks which are disseminated through portals like **“secureyourepc.in”, www.secureyoureelectronics.in and www.cert-in.org.in.**
- To create awareness about such frauds by using emails and sms, advisories are being issued by banks and telecom service providers which may be referred.

- Cross Border disputes.
- Timely Action on crime
- Encrypted content at ISP
- Awareness of cyber crimes and actions required.
- Implementation of regulations/policies regulating social media content (like posts, films, entertainment channels) or platform like youtube, facebook, whatsapp etc).
- Technical challenge in tracing the root of crime.

- Check if you've already been involved in a data breach.
- Check the strength of your passwords.
- Avoid common passwords.
- Trust no one (on emails)
- Secure your device.
- Think Before You Click
- Set up Two-factor or Multi-factor Authentication (MFA)
- Check CERT-In Updates on a Regular Basis.
- Keep Your Systems Updated
- Use Firewalls and Anti-viruses
- Avoid Online use of Debit Cards
- Learn About Phishing Attacks
- Avoid Unfamiliar Websites
- Avoid Useless Downloads
- Stay Cautious on Social Media
- Make Your Data Backup Regularly
- Don’t Use Public WiFi without a VPN

Thank You

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